MSC THESIS · HOCHSCHULE HEILBRONN · STATUS 11 NOV 2026

Can the L2 add-on make a small model reason better?

I built L2, a 19.6 M-parameter add-on that runs four internal "trajectories", lets them debate, and averages them, and attached it to the pretrained language model SmolLM2-135M. The aim was better reasoning on medical question answering. After nine L2 versions (v1–v5), three seeds per final model and three reasoning tests, **L2 has not shown better reasoning anywhere.** Its best gain turned out to be an answer preference. The positive results came from the data and from the evaluation method.

3 seeds per model (17, 23, 31) · paired bootstrap 95 % CI · * = the CI excludes 0

1 · The question

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4 · Data: old, evidence, pair loss

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Findings at a glance

WORKED

Training the decision from the evidence (T61)

PubMedQA 58.0 → 61.3 %* without any add-on. Following flipped evidence rises from 27 % to 79–80 %. 3-benchmark average +1.25*. Training uses a fifth of the text.

WORKED

Three reasoning tests

They showed that L2's +2-point PubMedQA gain was a preference for "yes" (+11.0* more yes answers), which accuracy alone hides.

NOT USEFUL

The L2 layer

Reasoning never shown. With the right data it adds −0.10 points. Text perplexity drops under 2 %, but the generated text is no better.

1 The question

Small language models answer medical questions mostly by pattern-matching. The thesis idea: add a separate "thinking" module on top of a normal language model. Inside it, several candidate versions of the model's internal state (**trajectories**) start from different angles. They read each other, revise, and agree on one answer state. This is meant to work like a small internal debate, all inside one forward pass, with no extra text generated.

The original design had two parts: L1 and L2. L1 is left for future work. This thesis tests **L2 alone**, on a pretrained model, against the same model without it.

> **Research question:** does SmolLM2 + L2 reason better on medical QA than SmolLM2 alone, trained on the same data? "Reason better" means more than higher accuracy (see section 6).

2 The model

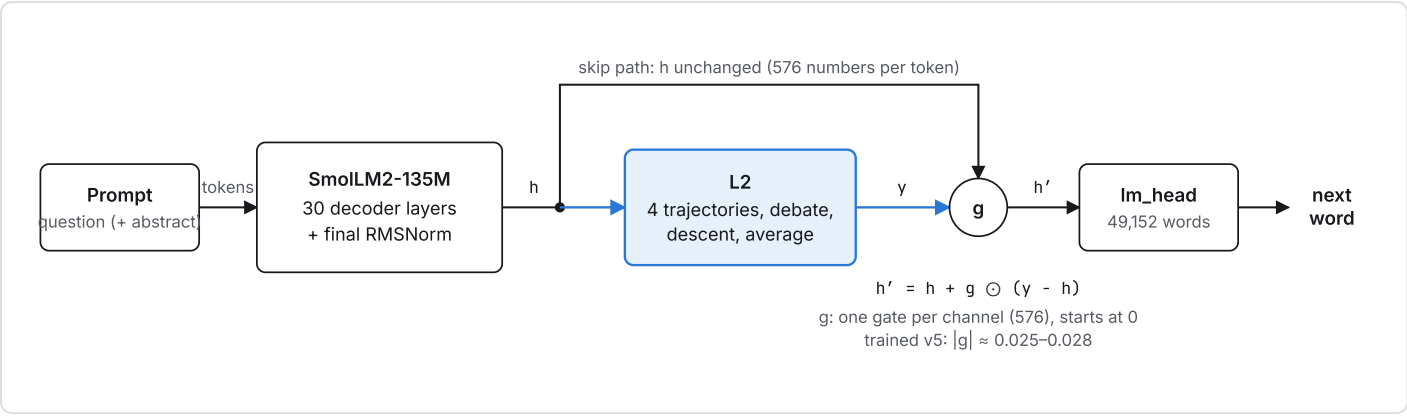
2.1 The backbone: SmolLM2-135M

SmolLM2-135M is a public pretrained decoder-only language model from Hugging Face. Its job is to predict the next word. We fine-tune all of its weights (full fine-tuning); LoRA was tried only as one variant (T60).

Property	Value
Parameters	134,515,584 (≈ 135 M)
Layers	30 decoder layers (self-attention + MLP each), RMSNorm, rotary position encoding
Hidden size	576 numbers per token (the "state" h)
Attention	9 query heads, 3 key/value heads (grouped-query attention)
Vocabulary	49,152 tokens; the output layer (lm_head) shares weights with the input embedding
Context used	up to 1,024 tokens per example (answers up to 512)

2.2 Where L2 attaches

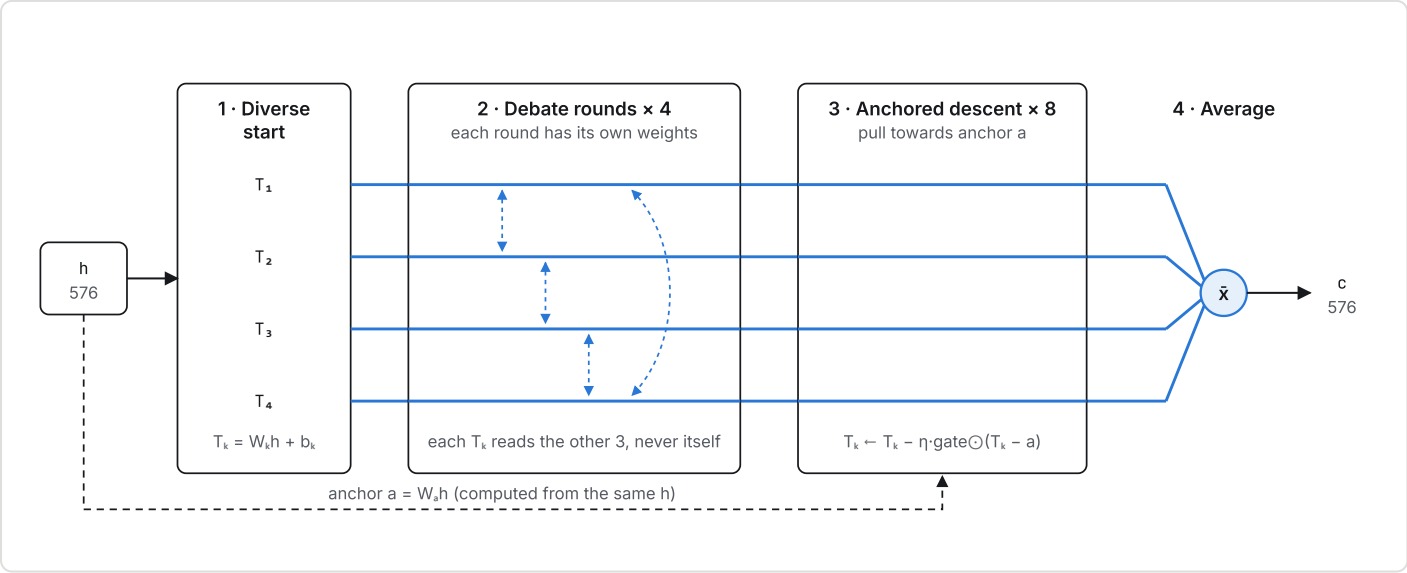
In the final model (v5), L2 sits between SmolLM2's last layer and its output layer. For every token, SmolLM2 produces its final state h (576 numbers). L2 computes a refined state y. A learned gate decides how much of y replaces h before the output layer turns it into next-word probabilities.



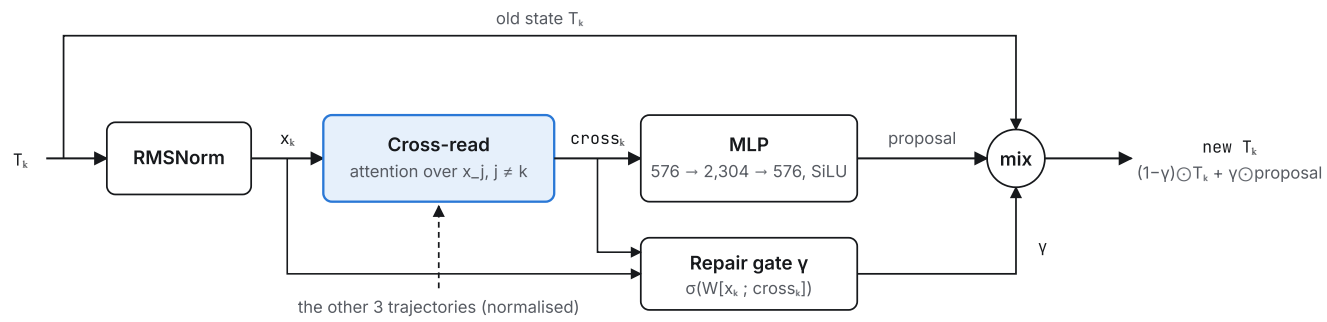
The whole model. At the start of training every gate is 0, so the model is exactly SmolLM2. Training can open the gate to mix in L2's output. In v5 the gates end at about 0.027: L2 changes roughly 3 % of the final state.

3 The L2 layer in detail

L2 works on one token at a time. It takes that token's final state `h` (576 numbers) and runs four steps: **diverse start → 4 debate rounds → 8 anchored descent steps → average**. All settings below are those of the final model v5 (`configs/v5_final.json`).



Inside L2, for one token. The blue lines are the four trajectories. The dashed blue arrows are the cross-reads in every debate round. The dashed black line is the anchor, computed from the same input h. The output c is L2's "consensus" state, which goes through RMSNorm and the gate (diagram above).



One debate round, for trajectory k. The repair gate γ lets a trajectory keep its own state rather than follow the others. Its bias starts at −2, so at first γ ≈ 0.12 and each round changes little.

3.1 Step by step

```
input   h            final state of one token from SmolLM2 (576 numbers)

1 start  T_k = W_k·h + b_k          k = 1..4
         W₁..W₄ start mutually orthogonal, with gains spread 0.8 → 1.25,
         so the four trajectories begin as four different "views" of h

2 debate (4 rounds, each with its own weights)
         x_k     = RMSNorm(T_k)
         cross_k = Σ_{j≠k} softmax_j( (Q x_k)·(K x_j) / √576 ) · V x_j
         γ_k     = sigmoid( W_γ [x_k ; cross_k] + b_γ )        b_γ starts at -2
         T_k    ← (1 - γ_k) ⊙ T_k + γ_k ⊙ MLP(cross_k)

3 descent (8 steps, η = 0.0625)
         a      = W_a·h                                        the anchor
         T_k    ← T_k - η · sigmoid(W_g [T_k ; a]) ⊙ (T_k - a)

4 average c  = (T₁ + T₂ + T₃ + T₄) / 4

output   y  = RMSNorm(c)
         h' = h + g ⊙ (y - h)      g = tanh(gate), 576 gates, start at 0
         next-word logits = lm_head(h')
```

3.2 Why each part is there

- **Diverse start.** If all four trajectories started equal, the debate would have nothing to argue about. Orthogonal projections make them start apart.

- **Debate.** Each trajectory looks at the others (attention over the other three, never itself) and proposes an update. The **repair gate** lets it decline the crowd, so a correct minority can survive a round.
- **Anchored descent.** This pulls every trajectory back towards an anchor computed from the original state, so the debate cannot drift away from what the model actually read. A second gate decides per channel how strongly to pull.
- **Average.** The consensus is the plain mean of the four (v5 uses `readout = mean`).
- **Spread floor (extra loss).** If the trajectories became identical, L2 would just be one MLP. The training loss therefore adds $\lambda \cdot \max(0, \rho_{min} - \rho)^2$, with ρ = trajectory variance / size of h, ρ_min = 0.05, λ = 1. This keeps the views apart.
- **Zero gate at start.** With g = 0, L2 starts as a no-op: both arms begin with exactly SmolLM2's output, so any difference is learned.

3.3 Parameter count (v5)

Part	Weights	Parameters
Diverse start	4 matrices 576×576 + 4 biases	1,329,408
One debate round	pre-norm 576 · cross-read Q, K, V 3×576² · repair gate 1,152→576 · MLP 576→2,304→576	4,317,120
4 debate rounds	4 × the above	17,268,480
Anchored descent	anchor 576² · descent gate 1,152→576	995,904
Output	output RMSNorm 576 · gate 576	1,152
L2 total		**19,594,944 (+14.6 %)**
SmolLM2 + L2		154,110,528

3.4 What L2 can and cannot see

> In v5, L2 works **per token**. It sees only that token's final state h; it does not read the other tokens or the abstract itself. Everything it knows about the question and the abstract is what SmolLM2 has already put into h. Because L2 sits at the very end, no later layer reads its output either. So L2 can only *reshape* what SmolLM2 has computed. It cannot gather new evidence. Later versions (v3–v4) gave L2 its own attention over the text, a verifier, a vote and placements inside the network. None of them did better than v5.

4 Data: the old data, the evidence data (T61), the pair loss (T63)

4.1 Training data (all public)

Dataset	What it is	Use
PubMedQA `pqa_artificial`	211,269 research questions, each with a PubMed abstract, the authors' conclusion, and an automatically generated yes / no label	Training, stage 1 (2,000 rows held out for checking)
MedQuAD	16,407 consumer health questions with answers from NIH websites (14,345 unique questions)	Split by unique question: one half for training (stage 2, 10 % of it for picking the checkpoint), the other half only as a benchmark
PubMedQA `pqa_labeled`	1,000 questions with expert yes / no / maybe labels	Benchmark only; it shares no PubMed id with the training set (checked)

4.2 The ”old data” (v1 – v5, T59, T60)

Stage 1 was trained as **question + abstract → write the authors' conclusion** (209,269 rows, about 85 M tokens per epoch). We then found two problems:

- **The labels are 92.8 % "yes":** 196,144 yes, 15,125 no, no maybe.
- **The yes / no word was never a training target.** The model only learned to write the conclusion. The PubMedQA benchmark scores exactly that yes / no word, so the benchmark answer was a side effect of conclusion writing.

This is why L2 could improve PubMedQA by learning to say "yes" more often (yes is the majority answer) without reading the evidence any better.

4.3 The evidence data (T61, `--evidence`)

This rebuilds stage 1 only; MedQuAD and every benchmark stay the same. The training answer now starts with the decision, and half of the rows force that decision to come from the evidence:

Rows	Prompt	Target	Count (seed 17)
Balanced	question + abstract	`yes. <conclusion>` or `no. <conclusion>`	29,934

Rows	Prompt	Target	Count (seed 17)
Flipped pairs	question + the conclusion only	`yes` / `no`	2 × 45,774

- **Balanced:** every "no" row, plus the same number of "yes" rows, so always saying "yes" scores only 50 %.
- **Flipped pairs:** each conclusion appears twice. Once as written, labelled with its real answer. Once with one negation added or removed (for example "does not improve" ↔ "does improve"), labelled with the opposite answer. Only a model that reads the sentence can get both right.
- Total: 121,482 short rows, about a fifth of the old text, so a training run takes about 0.9 h. The 1,000 benchmark questions are excluded.

4.4 The pair loss (T63, `--pair-loss 1.0`)

With plain cross-entropy, each row of a flipped pair is trained on its own. The literature (PairCFR, Qiu et al., ACL 2024) reports that counterfactual pairs help more when the loss compares both rows. So both rows of a pair are placed in the same batch, and the loss adds:

```
m         = logit(" yes") - logit(" no")      at the first answer token
loss_pair = W · softplus( m(row labelled no) - m(row labelled yes) ),   W = 1
          ≈ 0      when the answer flips with the evidence
          = log 2  (0.69) when both versions get the same answer
```

It reuses the logits the forward pass already computed: no extra compute. Both arms (with and without L2) get it.

5 Training

Item	Setting
Stage 1	PubMedQA (old data or evidence data), 1 epoch
Stage 2	MedQuAD training half: question → answer, 3 epochs; the better of best / last checkpoint on its 10 % slice
Batch	8 sequences × 4 accumulation steps = 32 per update; bf16 on one A100 80 GB per run
Learning rates	SmolLM2 5e-5 · L2 matrices 1e-4 · gates 1e-3; AdamW, cosine schedule per stage, 200 warm-up steps
Clipping	gradient norm 1.0, separately for SmolLM2 and L2

Item	Setting
Loss	next-token cross-entropy on the answer tokens + spread floor (L2 only) + pair loss (T63 only)
Fair comparison	same data, data order, split and starting weights for both arms; seeds 17, 23, 31

6 How we test

6.1 Multiple-choice accuracy

Benchmark	What it is	Questions	Chance
PubMedQA (`pqa_labeled`)	research question + abstract → yes / no / maybe; expert labels	1,000	33 %
MedMCQA (validation)	Indian medical entrance exams, 4 options	4,183	25 %
MedQA-USMLE (test)	US medical licensing exam, 4 options	1,273	25 %

Scoring follows lm-evaluation-harness: the question is asked in the training format (`Question: … Answer:`, with `Context:` for PubMedQA). The model's answer is the option with the highest mean log-probability per token. Nothing is generated, so no answer parsing is needed.

6.2 Text measures

- **Perplexity** of a reference text (lower = the model predicts it better): the MedQuAD benchmark half, MedMCQA expert explanations, PubMedQA conclusions.
- **Generated text:** the model writes its own MedMCQA explanation / PubMedQA conclusion (greedy decoding, 200 per set). We score ROUGE-L against the reference and the % of explanations that **name the right answer**.

6.3 The three reasoning tests (T59)

Test	What it measures	Reasoning would show as
1 · Where the gain comes from	PubMedQA accuracy split by the true answer, plus how often each answer is chosen	gains on yes *and* no questions, not just "says yes more often"
2 · Uses the abstract	% right with the abstract minus % right without it (same 1,000 questions). Never trained on	a starred gain: the answer depends more on the evidence

Test	What it measures	Reasoning would show as
3 · Follows flipped evidence	the authors' one-sentence conclusion is the only context, once as written and once with one negation flipped (164 questions). "Right on both" counts	a starred gain in "right on both"

> **Decision rule, fixed before the runs:** L2 "reasons better" only if tests 2 **and** 3 both show a starred gain over the same model without L2. From T61 on, flipped conclusions are part of the training data (other questions), so test 3 is no longer unseen; test 2 stays unseen.

6.4 Statistics

Every number is a mean over three seeds. The difference between two models comes with a **paired bootstrap 95 % confidence interval**: questions are resampled, the same resample is used in every seed, and both models are scored on the same questions. A star (*) means the interval excludes 0. Percentages of correct answers are reported as **percentage points**.

L2-off test: every L2 model is also scored with L2 switched off (gate = 0). If the score drops, the model depends on L2. If it stays, L2 is not doing the work at test time.

7 Versions and names

7.1 How we got here

Step	What we did	Result
v1	50 M-parameter model with L2, trained from scratch	Learned only the most frequent answer letter; moved to a pretrained backbone
v2.0 – v2.2	L2 on SmolLM2: at the end (v2.0), after layer 20 (v2.1), at several depths (v2.2)	At the end works best; small perplexity gains
v3, v3b, v3c	L2 reads the text itself, starts at the model's own state, adds a verifier, a vote, a confidence signal and a per-token gate; v3c "thinks twice"	No better than v2.2
v4	L2 inside the network and at the end, reading earlier layers	Worse than v2.2 (average −0.49*)
v5	Final L2 model = the v2.2 "end" recipe (section 3), three seeds	PubMedQA **+2.07***, average +0.72*, MedQuAD perplexity −0.16 %*

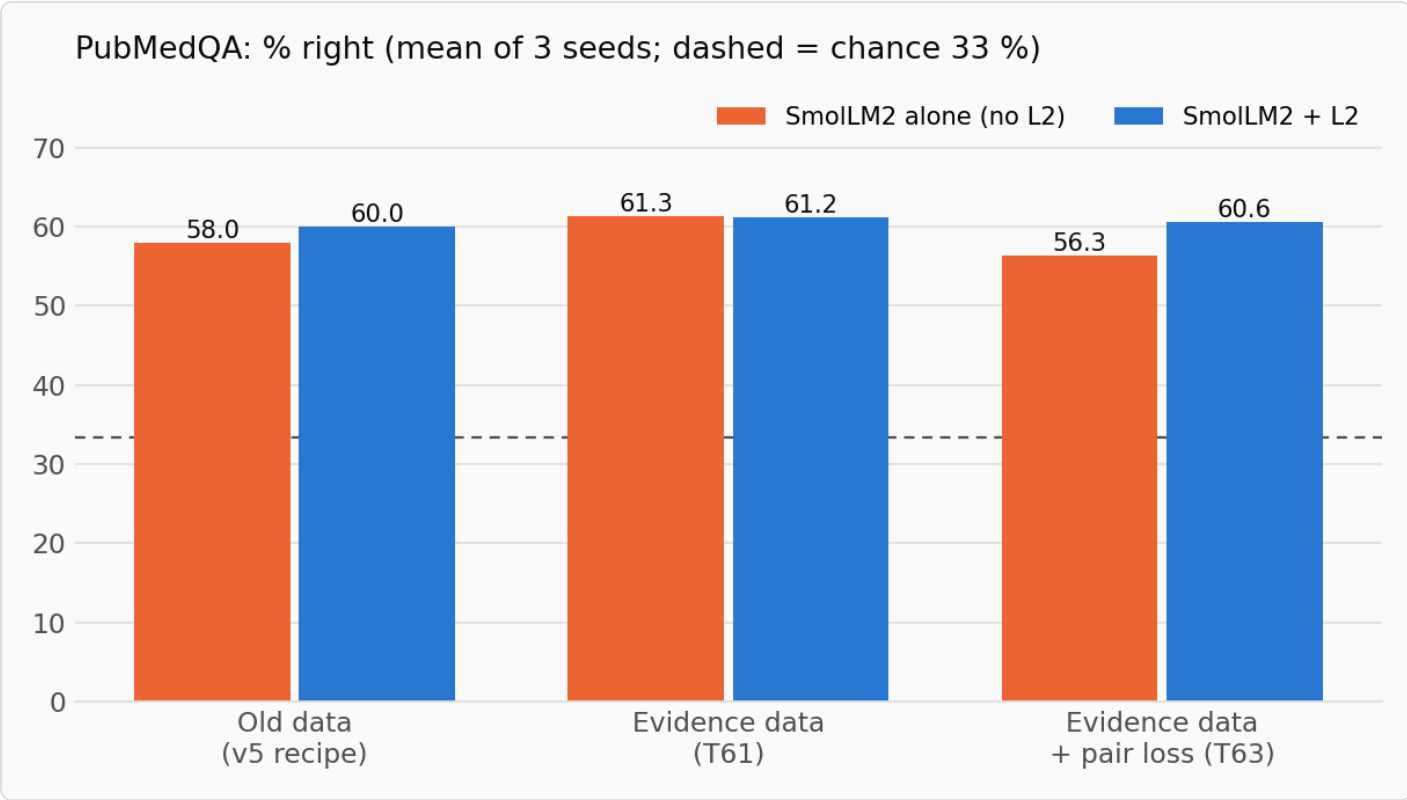
Step	What we did	Result
v5drop, v5out	L2 switched off on 30 % of training steps (T57); L2 output at the model's scale (T58)	Neither better than v5
T59	The three reasoning tests on v5	A preference: "yes" +11.0*; tests 2 and 3 not starred
T60	Train L2 first (warm-up), or LoRA instead of full fine-tuning	Best perplexity, but PubMedQA −15* (85 % "no"); LoRA answers "no" to everything
T61	Evidence data (section 4.3)	Both arms improve a lot; L2 adds nothing
T63	Pair loss (section 4.4)	Hurts the baseline; L2 model holds, but not through L2 at test time

7.2 Glossary

`T-numbers`	Entries in the project's issue list (`10_Known_Issues.md`). Each T is one problem or experiment; T59–T63 are the ones in this report.
`v-numbers`	Versions of the L2 design (v1 ... v5).
`baseline`	SmolLM2 alone (no L2), trained on the old data.
`v5`	SmolLM2 + L2, old data. Final L2 model.
`evbase / ev5`	SmolLM2 alone / SmolLM2 + L2, trained on the evidence data (T61).
`pairbase / pair5`	The same two, evidence data + pair loss (T63).
`gate g`	576 numbers deciding how much of L2's output reaches the output layer; 0 = L2 off.
`trajectory`	One of L2's four candidate versions of the token's state.
*	The 95 % confidence interval excludes 0.

8 Results

8.1 PubMedQA accuracy

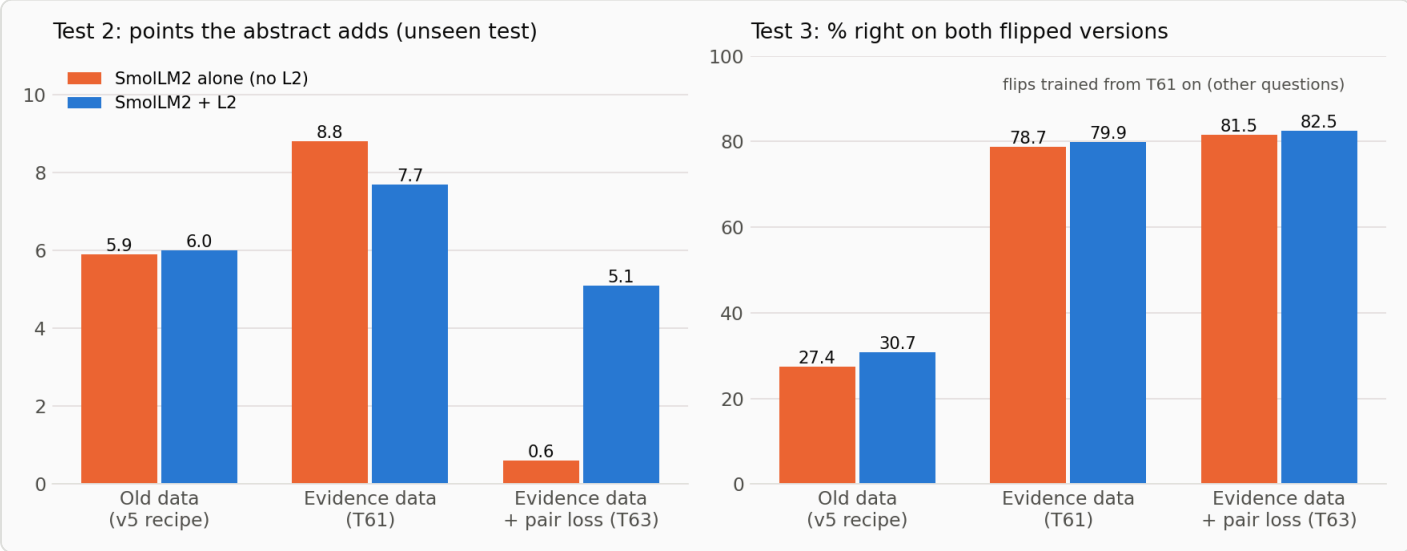


SmolLM2 alone · SmolLM2 + L2. Mean of three seeds; dashed line = chance.

Training data	SmolLM2 alone	SmolLM2 + L2	L2 minus alone [95 % CI]
Old data (baseline / v5)	58.0	60.0	**+2.07 [+0.57, +3.73]***
Evidence data (evbase / ev5)	61.3	61.2	−0.10 [−0.70, +0.47]
Evidence + pair loss (pairbase / pair5)	56.3	60.6	**+4.27 [+3.10, +5.57]***

- Against the old baseline (58.0), the evidence data gives **+3.30*** without L2 and +3.20* with L2. The 3-benchmark average is +1.25* / +1.22* (v5: +0.72*).
- MedMCQA and MedQA never change by more than ±0.5 points (no star). They stay near chance (25 %) for a 135 M model.

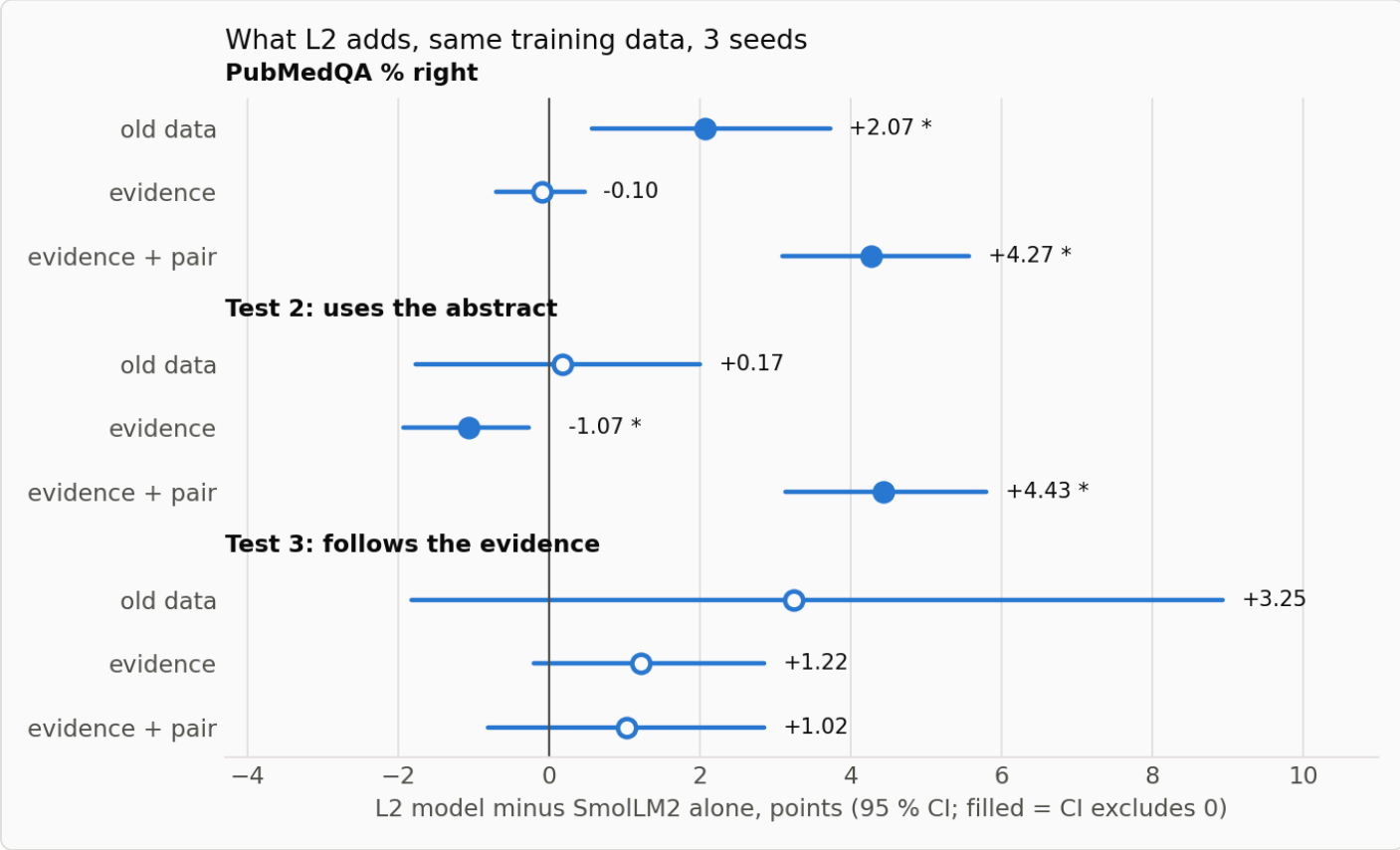
8.2 Reasoning tests



Test 3 jumps from about 27 % to about 80 % with the evidence data. This shows the training worked; it is no longer an unseen test. Test 2 stays unseen. The pair loss makes SmolLM2 alone almost stop using the abstract (0.6).

Model	Uses the abstract (test 2)	Follows flipped evidence (test 3)	Same answer on both versions
baseline / v5	5.9 / 6.0	27.4 % / 30.7 %	69.3 % / 65.7 %
evbase / ev5	8.8 / 7.7	78.7 % / 79.9 %	13.8 % / 12.2 %
pairbase / pair5	0.6 / 5.1	81.5 % / 82.5 %	14.0 % / 12.8 %

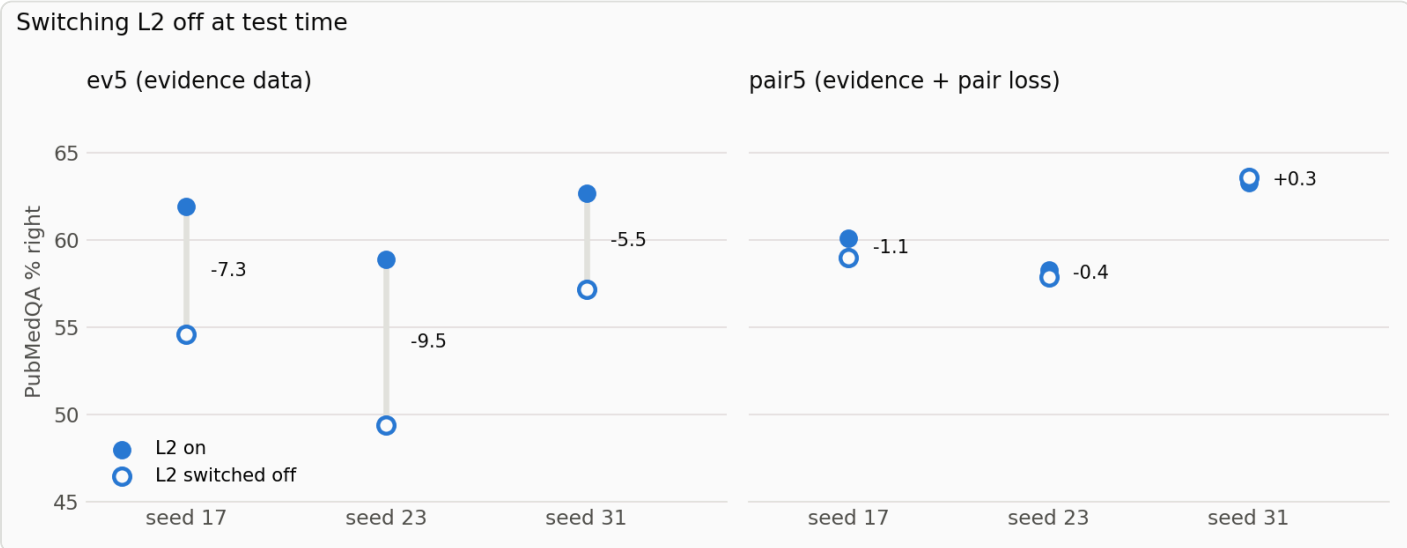
8.3 What L2 adds on the same data



Filled dot = the interval excludes 0. Test 3 is never starred, so the decision rule is never met.

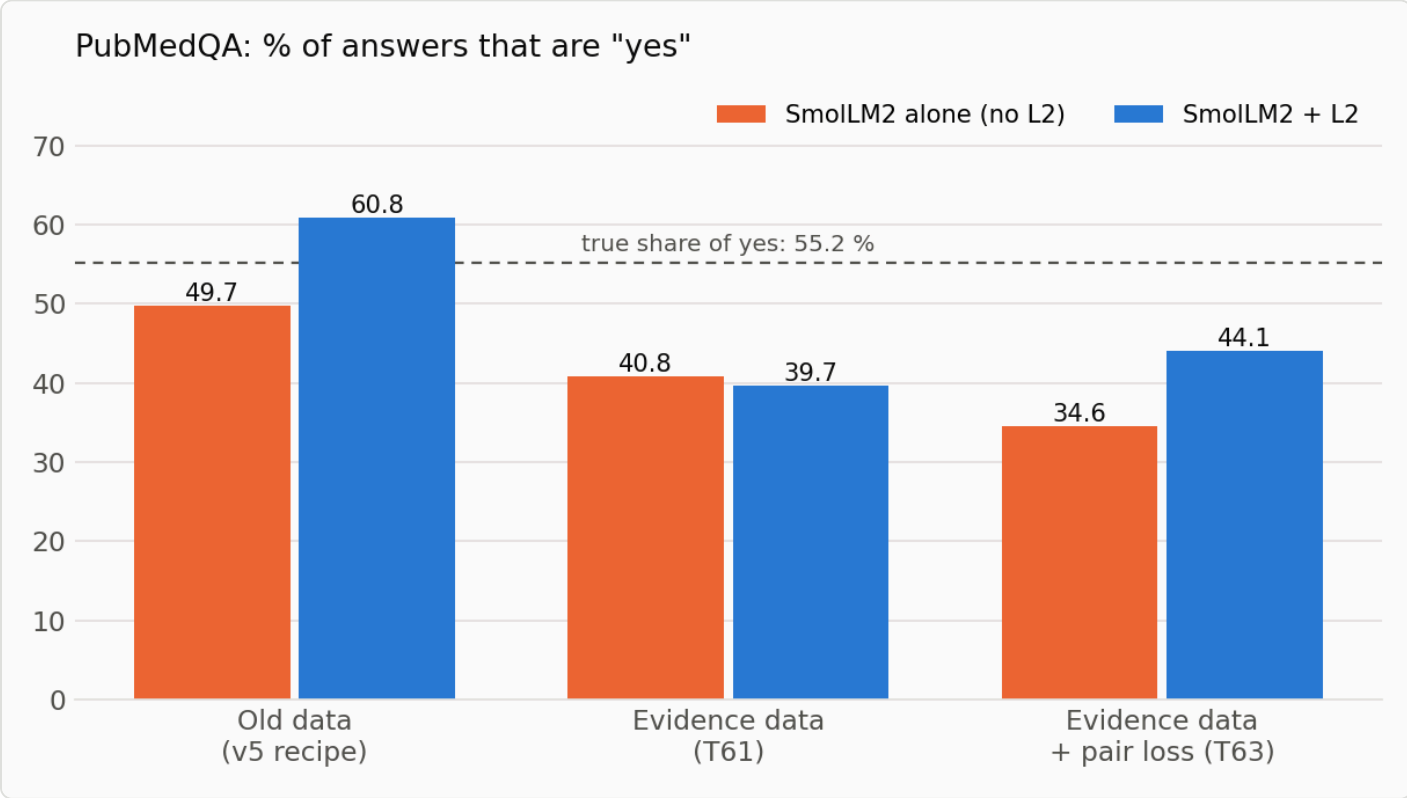
- **Old data:** the gain is a preference (+11.0* more "yes" answers); tests 2 and 3 are not starred.
- **Evidence data:** L2 adds nothing and uses the abstract slightly *less* (−1.07*).
- **Pair loss:** the first starred test-2 gain, on every seed (+4.8* / +5.8* / +2.7*). But it is measured against a baseline the pair loss damaged, test 3 is not starred, and the L2-off test (next) shows it is not L2's work.

8.4 Switching L2 off at test time



ev5 depends on L2 but is no better than the model trained without it: SmolLM2 handed part of its own work to L2. pair5 hardly uses L2 (gate 0.015 against 0.027). Its lead over pairbase sits in SmolLM2's own weights, so L2 changed how the backbone trained, not what is computed at test time.

8.5 Answer preference



True share of "yes": 55.2 %. v5 overshot (60.8 %). The evidence-trained models under-say "yes" (≈ 40 %): right on 87–88 % of the true-no questions and 57–58 % of the true-yes ones. The pair loss pushes SmolLM2 alone to 34.6 %.

8.6 Text generation

L2 minus alone, same data	Old data (v5)	Evidence data (ev5)
MedQuAD perplexity	**−0.16 %***	**−0.15 %***
MedMCQA explanation perplexity	**−1.78 %***	**−1.46 %***
PubMedQA conclusion perplexity	**−0.45 %***	**+0.84 %* (worse)**
Generated: explanation ROUGE-L	+0.05	+0.01
Generated: explanation names the right answer	−0.67	−0.83
Generated: conclusion ROUGE-L	−0.35	−0.02

- **L2 lowers perplexity a little** (under 2 %, on every seed), but the text it writes is no better: no generated-text measure differs significantly.

- The evidence data has the larger effect: explanation perplexity −5.33 %* without L2 and −6.72 %* with L2 against the old baseline. The cost is PubMedQA conclusion perplexity, +11.7 / +12.7 %* worse, because less conclusion text is trained.
- The pair loss makes generated explanations worse: ROUGE-L −1.99* (pairbase) and −3.15* (pair5) against evbase.

9 Verdict and conclusions

> **L2 is not useful for this task and is not worth keeping.** What the project produced comes from the work around it: the data finding and the three reasoning tests.

- **Reasoning:** never shown. The pre-registered test (starred gains on tests 2 and 3) was never passed.
- **Accuracy:** its only fair gain (+2.07* PubMedQA, old data) was a "yes" preference. With the evidence data it adds −0.10 points. MedMCQA and MedQA never moved.
- **Text generation:** perplexity drops by under 2 %, but the written text is no better.
- **Cost:** 19.6 M extra parameters (+14.6 %). The model learns to lean on L2 (ev5 loses 5.5–9.5 points with it off) without beating the model trained without it.
- **A likely reason (interpretation):** in v5, L2 only reshapes each token's final state (section 3.4). It has no access to evidence that SmolLM2 has not already encoded. The versions that let L2 read the text (v3–v4) were not better either.

What the thesis can claim

1. **The data finding.** The yes / no decision was never trained, and the data was 92.8 % "yes". Training the decision from balanced and flipped evidence improves PubMedQA (+3.3* points) and the 3-benchmark average (+1.25*), raises following flipped evidence from 27 % to 80 %, and is cheaper to train.
2. **The evaluation method.** Splitting accuracy by the true answer, "uses the abstract", the flipped-evidence test and the L2-off test turned an apparent +2-point "reasoning" gain into an answer preference. Accuracy alone would have hidden it.
3. **A careful negative result for L2**, measured against a same-data baseline on three seeds with confidence intervals.

10 For discussion

- Frame the thesis as a careful negative result for L2, with the data finding and the test method as the positive contributions?
- Stop experiments and start writing, or run one more check (for example an unseen flipped-evidence format, since test 3 is now trained-like)?
- Final model: proposed the evidence-data recipe **without** L2 (evbase). It scores the same as ev5 with 19.6 M fewer parameters.

Sources: `run.md` A2, A3o–A3s, A7k–A7n; reports `gain_*.md`, `reason_*.md` in `09-Code`; code `09-Code/src/ekdant/sandbox.py`, `model.py`, config `configs/v5_final.json`; design notes `07-`

Management/30_Design_v2_Pretrained_Backbone.md; issues T59–T63 in 07-Management/10_Known_Issues.md.